(a) Full prefill
reference behavior
p
d1
d2
d3
q
reference answer
(one causal pass)
(b) Composed cache
each di encoded after p only
p
d1
d2
d3
q
mismatched histories:
answer can drift
(c) Document-only repair
re-encode d2 alone
re-encoded
p
d1
d2
d3
q
d3 still misses d2:
drift persists
(d) SAFE-KV suffix repair
re-encode Suffix(s), s = 2
re-encoded
p
d1
d2
d3
q
consistent from s on:
answer restored
exact history
mismatched history
re-encoded
static prefix